

The empirical recurrence for OEIS A317462, recovered from its tail

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Abstract

OEIS A317462 records “Empirical recurrence of order 61” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 82$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 65$. Its last coefficient is $c_{61} = -4411392 \neq 0$, so no further tail satisfies anything shorter; any order-61 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A317462 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 2, 3, 4, 5, 6, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 61, 636, 16809, 386855, 9178195, 217053933, 5142922330, ...

The entry states:

Empirical recurrence of order 61 (see link above)

As of the “Last modified” line on the live entry (Jul 28 2018 21:00:14, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 82$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 82$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 61: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 61.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 164$ exact terms from $n = 4$ on, it returns order 61 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{61} c_i a(n-i),$$

c_1	18	c_{17}	−3439675009	c_{33}	10722493875671	c_{49}	65491675104
c_2	140	c_{18}	52828417367	c_{34}	9964209983169	c_{50}	137842433430
c_3	351	c_{19}	67367031671	c_{35}	−16419467259035	c_{51}	105659575920
c_4	−8369	c_{20}	−88487538102	c_{36}	−22812906349731	c_{52}	19788963608
c_5	−64057	c_{21}	−264473655196	c_{37}	8083926580728	c_{53}	−2603287696
c_6	−97868	c_{22}	−131612322888	c_{38}	30935438003344	c_{54}	11766493472
c_7	833196	c_{23}	345891858236	c_{39}	11227551185005	c_{55}	3982347200
c_8	5840536	c_{24}	703611050898	c_{40}	−16925723086256	c_{56}	1666655232
c_9	7788795	c_{25}	619100372390	c_{41}	−17144925022971	c_{57}	−271510272
c_{10}	−33871333	c_{26}	−557203440868	c_{42}	426486481149	c_{58}	440728064
c_{11}	−172923617	c_{27}	−2737602450391	c_{43}	6774908143714	c_{59}	117145600
c_{12}	−128074676	c_{28}	−1607029698577	c_{44}	2369450117947	c_{60}	44281856
c_{13}	1056237876	c_{29}	4390570032566	c_{45}	−1005164546290	c_{61}	−4411392
c_{14}	2188439428	c_{30}	4175932266898	c_{46}	−498401514342		
c_{15}	−2794150470	c_{31}	−5958343308112	c_{47}	70563538191		
c_{16}	−13617251169	c_{32}	−5139893132786	c_{48}	−165490408290		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 65$, and it is the recurrence OEIS A317462 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{61} - c_1 x^{61-1} - \dots - c_{61}$. Its constant term is $-c_{61} = 4411392$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 61 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 61, so $\deg q = 61$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 61, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 61, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -4411392 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-61 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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